

Image Restoration Report

Goal: generate 0/1 noise masks, restore missing pixels, reduce L2 error, and improve SSIM/Cosine similarity.

Implementation: main.ipynb implements `noise_mask_image()` and `restore_image()` without changing function signatures or return types.

Environment: conda torch environment with numpy, opencv-python, scipy, and matplotlib.

Noise mask: for each channel and each row, exactly $\text{round}(\text{width} * \text{ratio})$ pixels are randomly set to zero.

Restoration: OpenCV Telea inpaint provides a strong candidate image, then local binary linear ridge regression fits $\text{value} = a*x + b*y + c$.

Adaptive windows: the radius starts from size and expands to $2*\text{size}$ and $4*\text{size}$ if local valid samples are insufficient.

Fallbacks: local mean and channel global mean are used when regression has too few valid samples.

Fusion: $\text{missing pixels} = 0.55 * \text{inpaint} + 0.35 * \text{local_regression} + 0.10 * \text{local_mean}$. Known pixels remain unchanged.

A.png restoration, noise ratio = [0.8, 0.4, 0.6]

Original



Degraded



Restored



Metric comparison: restoration improves all tested cases

case	L2 bad	L2 restore	Cos bad	Cos restore	SSIM bad	SSIM restore
A-804060	933.0	54.3	0.6275	0.9990	0.1624	0.9908
A-406080	921.3	54.6	0.6394	0.9990	0.1820	0.9907
forest	252.2	66.5	0.6321	0.9789	0.2105	0.8438
mona	253.9	32.2	0.6323	0.9952	0.2937	0.9848
potala	259.7	59.0	0.6325	0.9844	0.2636	0.9333
athens	391.5	70.0	0.6324	0.9903	0.2279	0.9447
xihu	407.9	28.6	0.6324	0.9985	0.2725	0.9946

Conclusion and Submission Notes

Average L2 reduction: 85.94%.

Average Cosine gain: 0.3595.

Average SSIM gain: 0.7243.

The method is stable for high-ratio random masks, keeps observed pixels fixed, and uses local image structure to recover missing regions.

The target functions are self-contained in main.ipynb, so the submission extractor can run them with the import cell only.

Generated images are in results/. The metric table is in results/metrics_summary.csv.